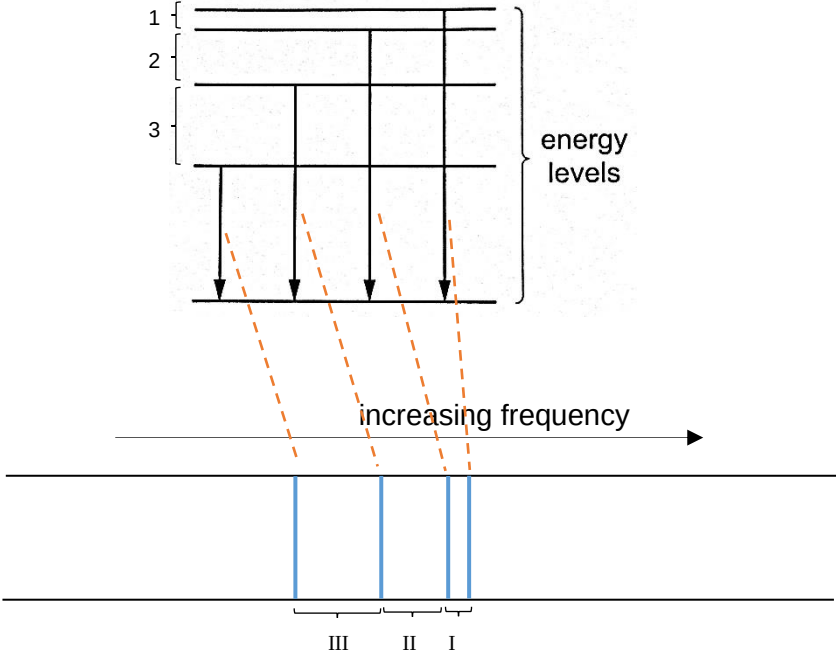


DUNMAN HIGH SCHOOL
2023 H2 PHYSICS (YEAR 6)

9	(a)	discrete quantity / packet / quantum of energy of electromagnetic radiation energy of photon = Planck constant $\times$ frequency	B1 B1
	(b)	- existence of threshold frequency for incident light - rate of emission of electrons is proportional to intensity of incident light - max. kinetic energy of electron dependent on frequency of incident light - max. kinetic energy independent of intensity of incident light (any three, 1 each, max 3)	B1 B1 B1 B1
	(c)	either $E = hc/\lambda$ $= hc/(450 \text{ nm})$ $= 4.4 \times 10^{-19} \text{ J}$ or 2.8 eV as $2.8 \text{ eV} < 3.5 \text{ eV}$, $\Rightarrow$ no emission or $\Phi = hc/\lambda_0$ $\Rightarrow 3.5 \text{ eV} = hc/\lambda_0$ $\Rightarrow \lambda_0 = 355 \text{ nm}$ as $355 \text{ nm} < 450 \text{ nm}$, $\Rightarrow$ no emission	C1 M1 A1
	OR	or $\Phi = hf_0$ $\Rightarrow 3.5 \text{ eV} = hf_0$ $\Rightarrow f_0 = 8.45 \times 10^{14} \text{ Hz}$ $450 \text{ nm} \rightarrow 6.67 \times 10^{14} \text{ Hz}$ as $6.67 \times 10^{14} \text{ Hz} < 8.45 \times 10^{14} \text{ Hz} \Rightarrow$ no emission	C1 M1 A1
	(d)(i)	when electrons/atoms are in their lowest energy/minimum energy/most stable (state)	B1
	(d)(ii)	excitation promotes an electron to a higher energy level. ionization occurs when an electron receives enough energy to leave the atom.	B1 B1
	(e)	electrons occupy <u>discrete</u> energy levels / difference in energy levels are discrete. energy of photon is proportional to frequency ($E = hf$) atom/electron can be excited only when energy of photon exactly matches the difference in energy levels (1 to 1 interaction / all energy of photon absorbed)	B1 B1 B1

	(f)	Four lines drawn. Correct spacings:  (since the energy levels are drawn to scale, the spacings III, II and I should correspond to the spacings 3, 2 and 1, respectively)	M1 A1
	(g)(i)	Δx decreases and Δp increases	B1
	(g)(ii)	link between momentum p and kinetic energy E_k: $\frac{p^2}{2m} = E_k$ increased uncertainty in momentum implies increased uncertainty in kinetic energy	B1 B1
	(g)(iii)	For very small orbits the uncertainty in the kinetic energy becomes large enough (to make the total energy positive) that the electron will escape	B1